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1 What Comparison Means

There are several ways to think about the differences between two files. One way to think of the differences is as a series of lines that were deleted from, inserted in, or changed in one file to produce the other file. `diff` compares two files line by line, finds groups of lines that differ, and reports each group of differing lines. It can report the differing lines in several formats, which have different purposes.

GNU `diff` can show whether files are different without detailing the differences. It also provides ways to suppress certain kinds of differences that are not important to you. Most commonly, such differences are changes in the amount of white space between words or lines. `diff` also provides ways to suppress differences in alphabetic case or in lines that match a regular expression that you provide. These options can accumulate; for example, you can ignore changes in both white space and alphabetic case.

Another way to think of the differences between two files is as a sequence of pairs of bytes that can be either identical or different. `cmp` reports the differences between two files byte by byte, instead of line by line. As a result, it is often more useful than `diff` for comparing binary files. For text files, `cmp` is useful mainly when you want to know only whether two files are identical, or whether one file is a prefix of the other.

To illustrate the effect that considering changes byte by byte can have compared with considering them line by line, think of what happens if a single newline character is added to the beginning of a file. If that file is then compared with an otherwise identical file that lacks the newline at the beginning, `diff` will report that a blank line has been added to the file, while `cmp` will report that almost every byte of the two files differs.

`diff3` normally compares three input files line by line, finds groups of lines that differ, and reports each group of differing lines. Its output is designed to make it easy to inspect two different sets of changes to the same file.

These commands compare input files without necessarily reading them. For example, if `diff` is asked simply to report whether two files differ, and it discovers that the files have different sizes, it need not read them to do its job.

- [Hunks](#)
- [Suppressing Differences in Blank and Tab Spacing](#)
- [Suppressing Differences Whose Lines Are All Blank](#)
- [Suppressing Differences Whose Lines All Match a Regular Expression](#)
- [Suppressing Case Differences](#)

- [Summarizing Which Files Differ](#)
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